EUCLID'S ALGORITHM ON ONE PAGE

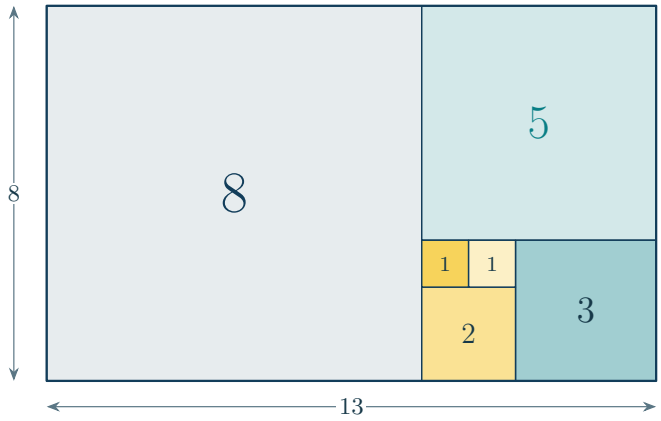
Cutting squares from a rectangle

From 13×8 to common factors, continued fractions and the golden ratio.

1 / THE PICTURE IS A DIVISION

Remove the biggest square, then repeat on the remainder

Begin with a 13×8 rectangle. Cut parallel to its edges and repeat on the rectangle left over.



Square side lengths: $8, 5, 3, 2, 1, 1$.

Division is repeated square cutting.

$$\begin{aligned} 13 &= \mathbf{1} \times 8 + 5, \\ 8 &= \mathbf{1} \times 5 + 3, \\ 5 &= \mathbf{1} \times 3 + 2, \\ 3 &= \mathbf{1} \times 2 + 1, \\ 2 &= \mathbf{2} \times 1 + 0. \end{aligned}$$

The quotients $1, 1, 1, 1, 2$ count the squares at each size: one of each, then two unit squares.

$$\mathrm{HCF}(13, 8) = 1$$

The last non-zero remainder.

2 / WHY THE ALGORITHM WORKS

The common factor survives every cut

Write $\gcd$ for the highest common factor. If $a = qb + r$, any common divisor of a, b divides $r = a - qb$. Conversely, any common divisor of b, r divides $a = qb + r$. Thus

$$\boxed{\gcd(a, b) = \gcd(b, r)}.$$

For our rectangle,

$$\begin{aligned} \gcd(13, 8) &= \gcd(8, 5) = \cdots \\ &= \gcd(1, 0) = 1. \end{aligned}$$

Geometrically, 1 is the largest square side for a regular grid tiling of the whole rectangle. Doubling every length gives a 26×16 rectangle and $\gcd(26, 16) = 2$.

3 / READ THE QUOTIENTS

A continued fraction

Turn the remainder fraction into a reciprocal:

$$\frac{13}{8} = 1 + \frac{5}{8} = 1 + \frac{1}{8/5}.$$

Repeat inside the denominator:

$$\frac{13}{8} = 1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{2}}}} = [1; 1, 1, 1, 2].$$

The bracket lists the five quotients. Since $2 = 1 + 1/1$, the same number can also be written $[1; 1, 1, 1, 1, 1]$: a finite fraction containing only ones.

4 / RUN THE CONSTRUCTION BACKWARDS

Fibonacci numbers build the rectangle

Read the square sides in reverse, $1, 1, 2, 3, 5, 8, 13, \ldots$, and each new term is the sum of the preceding two:

$$F_{n+1} = F_n + F_{n-1}, \qquad F_1 = F_2 = 1.$$

Starting from a unit square, attach a square along the longer side at each step. A rectangle with sides a, b becomes one with sides $a + b, a$, so its aspect ratio changes by

$$x = \frac{a}{b} \quad \longmapsto \quad 1 + \frac{1}{x}.$$

The successive ratios converge:

$$1,\ 2,\ \frac{3}{2},\ \frac{5}{3},\ \frac{8}{5},\ \frac{13}{8},\ \frac{21}{13}, \ldots \longrightarrow \phi.$$

5 / A SHAPE THAT REPEATS

The golden rectangle

Remove a unit square from an $x \times 1$ rectangle, $1 < x < 2$. The remainder is $1 \times (x - 1)$. For the same shape after a quarter-turn,

$$x = \frac{1}{x - 1} \quad \Longleftrightarrow \quad x = 1 + \frac{1}{x}.$$

$$\phi^2 - \phi - 1 = 0, \qquad \phi = \frac{1 + \sqrt{5}}{2}$$

$$\phi = [1; 1, 1, 1, \ldots] \approx 1.618034$$

Our $13/8 = 1.625$ is a rational approximation. A true golden rectangle repeats the cutting process indefinitely: each remainder is a smaller copy of the original.

Companion: squared rectangles. **Explore:** NRICH: Euclid and continued fractions; NRICH: Fibonacci and the golden ratio.

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